

MINI PROJECT

Automatic Street Light System

Using IR Sensor

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Introduction

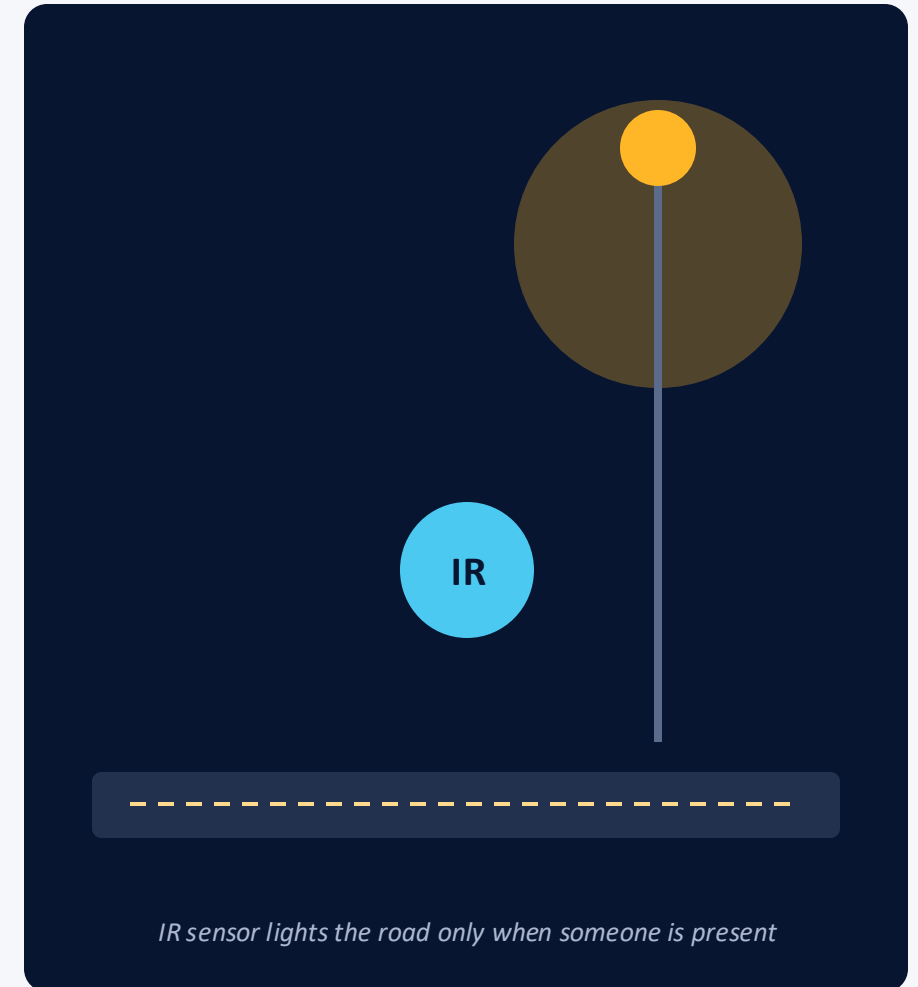
The Problem

Conventional street lights are switched ON/OFF manually or on fixed timers. This wastes electricity when roads are already lit by daylight, and lights stay on all night even when no one is around.

Our Solution

A smart street light that uses an IR sensor to detect vehicles or pedestrians on the road — turning the light ON automatically only when something is present, and OFF when the road is clear.

Result: an energy-efficient, self-operating lighting system with no manual intervention.



Objectives

What we set out to achieve with this project

01

Automate street lighting

Turn lights ON/OFF automatically based on IR sensor input, without human effort.

02

Save energy

Avoid unnecessary power usage whenever no one is on the road.

03

Detect presence

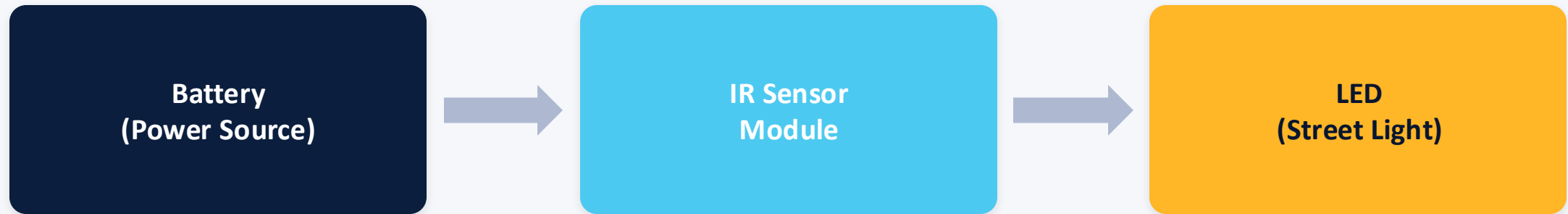
Use an IR sensor to sense vehicles or pedestrians and light up only on demand.

04

Build a low-cost prototype

Design a working model with easily available, inexpensive components.

System Block Diagram



Signal flow: The battery powers the IR sensor module → the module continuously monitors the road for vehicles or pedestrians → its output pin directly drives the LED, turning it ON when an object is detected and OFF when the road is clear. It's a simple, direct circuit — no relay, transistor, or microcontroller required.

Components Used



IR Sensor Module — detects vehicles or pedestrians and gives a direct output signal.



Battery — power source for the sensor module and LED.



LED — acts as the street light, wired straight to the sensor's output.



Resistor — limits current to protect the LED.



Connecting Wires — link the battery, sensor, and LED together.

Just 5 components, wired directly — battery in, LED out, IR sensor decides when.

Working Principle



Presence Detection (IR)

The IR transmitter continuously emits infrared light across the road. When a vehicle or pedestrian passes, the IR receiver detects the reflected beam and its output pin changes state immediately.

The sensor's output pin is wired directly to the LED. When it goes HIGH (object detected), current flows straight through the LED and it lights up instantly — no relay, transistor, or microcontroller in between.

Decision Logic

No object

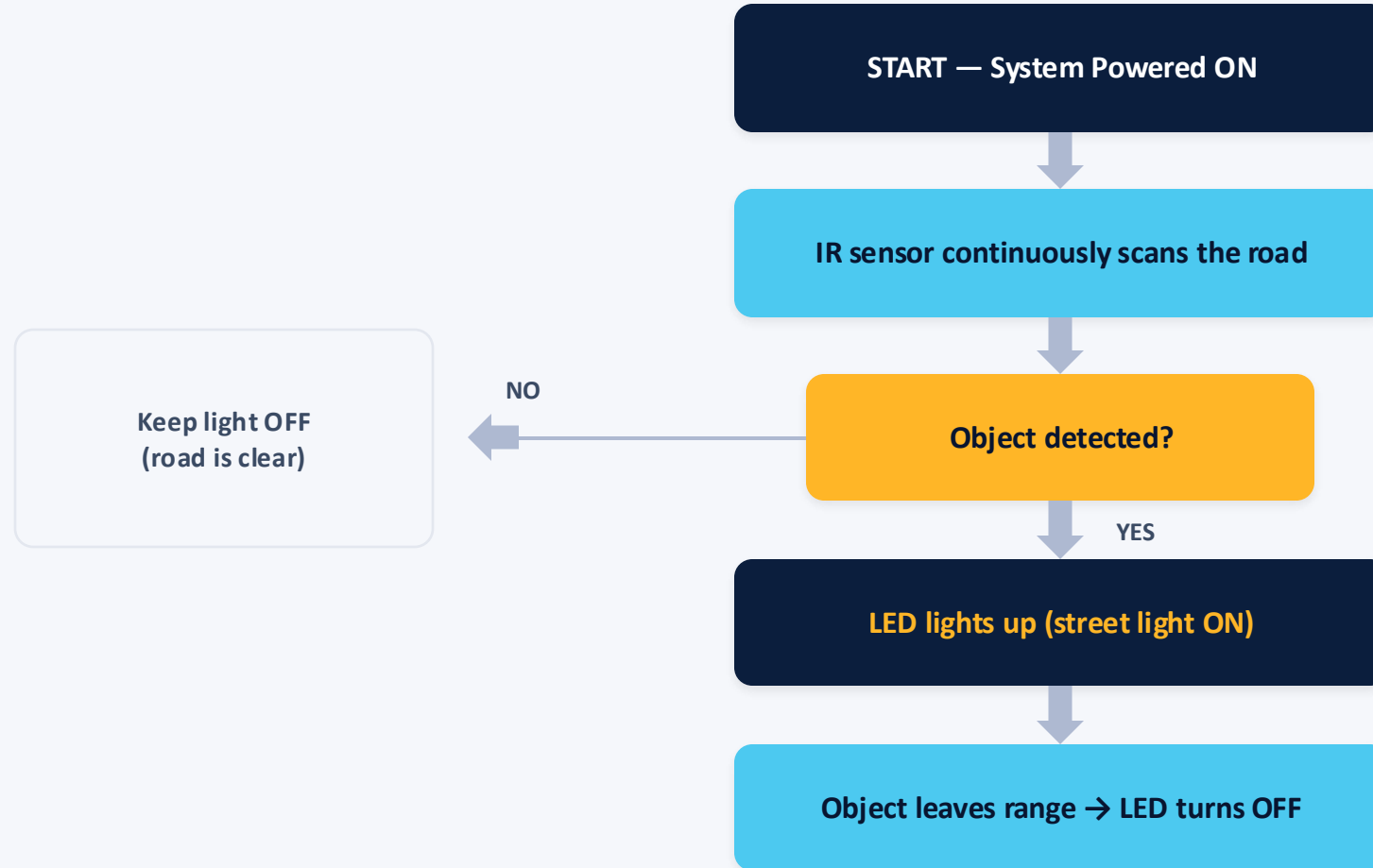
LED stays OFF

Object detected

LED turns ON

The LED stays off by default and lights up immediately whenever the IR sensor detects a vehicle or pedestrian — a direct, single-step connection.

Operation Flowchart



Advantages



Energy Saving

Lights only when someone is actually on the road — cuts electricity consumption significantly.



Fully Automatic

No manual switching or fixed timers; the system adapts to real conditions.



Longer Bulb Life

Reduced ON-time means less wear, extending the lifespan of the lights.



Cost Effective

Built from low-cost, easily available components — low setup cost.



Low Maintenance

Just a battery, sensor, and LED — no relay or microcontroller to fail.



Eco-Friendly

Lower power usage directly reduces the carbon footprint.

Applications

1

City Streets & Highways

2

College / School Campuses

3

Residential Colonies

4

Parking Lots

5

Parks & Walkways

6

Industrial Premises

Conclusion

The Automatic Street Light System successfully uses an IR sensor to switch street lighting on only when a vehicle or pedestrian is present. This delivers meaningful energy savings over always-on or timer-based lighting, while remaining low-cost, simple to build, and easy to maintain.

Future Scope

- IoT integration for remote monitoring & control
- Solar-powered supply for full energy independence
- Auto-dimming based on traffic density



Thank You

Questions & Suggestions Welcome

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